

Clausal parentheticals in disguise: the borrowing of *which* in code-mixing speech in Hong Kong Cantonese*

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Abstract. This paper discusses an understudied construction in code-mixing speech in Hong Kong Cantonese, whose signature feature is the use of the English relative pronoun *which* (*which*-constructions). These constructions apparently constitute a relative construction that introduces an atypical, post-nominal modification of the antecedent in Cantonese. While they have been characterized as an instance of lexico-syntactic transference or structural borrowing of post-nominal modification, I argue against such a modification account, and propose that *which*-constructions are best regarded as clausal parentheticals. I suggest that (i) they involve no syntactic borrowing of modification structures, and that (ii) the relative pronoun *which* is best regarded as an overt Force operator that establishes a predication relation between a null Topic and the rest of the clause. It is lexically borrowed from English to introduce parenthetical meanings. This study contributes our understanding of parenthetical meanings as well as borrowing in code-mixing speech in Hong Kong Cantonese.

Keywords: borrowing, code-switching speech, parentheticals, relative constructions, force operator, Hong Kong Cantonese

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1. Introduction

Language contact usually leads to borrowing in a language. Under the multilingual environments, Hong Kong Cantonese (hereafter Cantonese) is no exception. Researches on code-mixing of Cantonese and English in Hong Kong has emerged since 1970s (see, for overviews, Gibbons 1987, Li 1999, i.a.). In addition to a significant amount of studies on lexical borrowing, a less discussed aspect of borrowing is whether syntactic/structural borrowing is also attested in Cantonese. A case of particular interest concerns utterances like (1), as noticed in Chan (1993), T. C. Leung (2001), and K. W. Leung (2010). The English relative pronoun *which* is used post-nominally to introduce a clause, in a way similar to English relative constructions at first glance. In particular, the gap in the bracketed clause (represented by Δ) seems to indicate the position where the corresponding head (if any, boxed) is interpreted in the clause. Since the signature property of these clausal structures is the usage of *which*, I refer to them as ***which*-constructions**.

(1) Examples of *which*-constructions

a. Ngo m-tungji keoi ge jigin, [**which** Δ does not mean ngo zang keoi].
I not-agree 3SG GE opinion WHICH does not mean I hate 3SG
'I do not agree to his opinions, which does not mean I hate him.' (Chan (1993), p.9)

b. Keoi gin-dou bou dinwaa [**which** keoi waa soeng maai Δ] wo5.
3SG see-ASP CL telephone WHICH 3SG say want buy SFP
'He has seen a telephone, which he says he wants to buy.' (T. C. Leung 2001, p.58)

Which-constructions represent an interesting case in the study of code-switching, as they are substantially different from how modification is positioned in Cantonese, which is strictly pre-nominal, as given in (2).

(2) Pre-nominal modification in Cantonese

- a. [liksi (ge)] gaaufo syu
history GE textbook 'History textbooks'
- b. [Ginhong ge] sailou
healthy GE kid 'Heathly kids'
- c. [Ngo hou zungji ge] syu
I very like GE book 'The books that I like'

Earlier characterizations suggest that *which*-constructions involve a post-modifying structure (e.g., Chan 1993). K. W. Leung (2010) argues that "[t]he relative construction follows English grammar, forming a post-modifying relative clause with an English relative pronoun *which* preposed to the beginning of the clause" (p.63). It has thus been taken as an instance of *lexico-syntactic transference* (Li 1999; K. W. Leung 2010) or *constructional borrowing* (Chan 2022). Particularly, the constructions are thought to involve both (i) the lexical borrowing of the relative pronoun *which*, and (ii) the syntactic borrowing of the post-nominal, modifying relative structures of English.

The goal of this paper is two-fold. First, I argue that *which*-constructions should not be regarded as a case of syntactic/constructional borrowing of modification from English; instead, they merely instantiate a case of lexical borrowing of *which*. Second, and relatedly, I show that *which*-constructions show no quality of being a post-nominal modifier. Rather, they are best regarded as *clausal parentheticals*, a standalone clause with a distinct illocutionary force from the host clause. The expression *which* is lexically borrowed and serves as an overt force operator in the sense of Koev (2022).

To develop an understanding of *which*-constructions, I briefly trace the discussions on *which*-constructions in the literature in section 2, and then focus on the following two properties of *which*-constructions, namely, (i) whether it qualifies as a relative construction that contributes modification (section 3), and (ii) the nature of the propositional meaning introduced in *which*-constructions (section 4). I propose a clausal parenthetical analysis on *which*-constructions in section 5. In section 6, I provide further evidence for a clausal parenthetical approach to *which*-constructions and conclude that there is no syntactic or constructional borrowing in Cantonese code-mixing speech in section 7.

2. Backgrounds on study of *which*-constructions

Which-constructions receive very limited attention in the literature, but examples have been noticed no later than 1990s, as shown in (1a). K. W. Leung (2010) is the first attempt to document *which*-constructions. He collected 20 spontaneous/naturalistic examples in a diary kept for three months. He also ran a small judgment survey on 8 instances of *which*-constructions. The sentences are judged as 2.24/4 by 22 college students. Speaker variations towards the acceptability of *which*-constructions are reported, and they are most acceptable among college students. As he reports, the sentences may be “rated with low scores in acceptability judgment task even by participants who use this construction” (*ibid.*, p.23). This may be due to the conformity to a prescriptive norm. His study also reveals that *which*-constructions are attested in both written and spoken forms.

Recent internet searches reveal that the prevalence of *which*-constructions are underestimated. I collected more than 200 instances are collected in Nov-Dec 2022, from the internet of different sources, including forums, social network services, blogs, interviews, etc. Two examples are given below in (3).

(3) *Which*-constructions in naturalistic contexts

- a. Sailou faangung zau jau munzukgam [**which** keoidei fannhok jatzik
Kid work then have satisfaction WHICH 3PL study straight
dak-m-dou Δ].
gain-not-able

‘The young people at work usually get a sense of satisfaction, which they don’t get any at school.’

- b. Keoi zihau sau-dou dai-jat fung seon, [**which** Δ hai seonjungkaat
3SG then receive first CL letter WHICH be credit.card
gongsi gei Δ lei jiu keoi waan cin].
company send come request 3SG pay.back money

‘Then s/he received the first letter, which is sent from the credit card company to request payment.’

Note that most of the reported data in this paper are based on these instances, with slight modifications in some cases. Unacceptable cases are based on five native speakers who self-identify themselves as users of *which*-constructions.

Two remarks are in order. First, while the rest of this paper focuses on *which*-constructions in Cantonese, they are also fairly common in colloquial written texts of Mandarin, as exemplified in (4), although they have not been systematically documented in Mandarin so far. According to the native Mandarin speakers I consulted, they are judged highly marginal when uttered in speech. This suggests that the incorporation of *which*-constructions into Chinese languages is at different stages different: they are acceptable in both written and spoken forms in Cantonese, but only in written forms in Mandarin.

(4) Which-constructions in Mandarin

a. Changjian de zhiyou Yuenanfen, [**which** wo bu xihuan].
common DE only Pho WHICH I not like
‘Only Pho is common, which I do not like.’

b. Ruguo Allen you shi ge zinu, [**which** ta meiyou], na-me...
if Allen have 10 CL children WHICH he not.have then
‘If Allen had 10 children (he didn’t), then ...’

Second, while most cases of *which*-constructions are sentence-final, medial positions are possible for *which*-constructions, a position suggested to be unattested in K. W. Leung (2010). In (5), the *which*-RC can be inserted in the middle of the sentence. There is no general ban on the medial position.

(5) Sentence-medial *which*-constructions

Janwai haa-ci [**which** jatding wui jau haa-ci], moujan zi zung
because next-time WHICH for.sure will have next-time no.one know still
wui-m-wui gam gaandan.
will-not-will so simple

(Lit.) ‘Because next time, which (I am) sure that there will be next time, no one knows if it will still be simple.’

Before I move on to the next section, it is instructive to fix some terminology when describing *which*-constructions. The configuration of *which*-constructions can be represented in (6), where they typically follow their host clause. As we will see shortly, the gap (Δ) is not obligatory in the construction. But if there is one, it usually corresponds to a noun in the host clause that appears to be modified by the *which*-construction (e.g., (1b) and (3), boxed elements), but its presence is not required (see (1a)). For convenience, I will refer to the corresponding element as a head, if it is identifiable in the host clause.

(6) A typical configuration of *which*-constructions

[Host clause ... (X_{head}) ...], [*which*-construction **which** ... (Δ) ...].

3. *Which*-constructions ≠ English relative constructions

Despite surface similarities, *which*-constructions exhibit a number of differences compared to relative constructions in English. I argue that the two should not be conflated, as the former has the syntax of a root clause and shows no syntactic dependencies with any element in the host clause.

3.1. *Which* and *which* only

First of all, the choice of relative pronoun is not grammatically constrained by the head in *which*-constructions. Unlike English relative clauses, even the head refers to human beings, the use of *which* is allowed.

(7) No dependency between *which* and the head

Keoidei jiu ceotsin dak B/C loeng-zou honang dai-fan gwo [keoidei]
they need promote only B/C two-group may low-mark than them,
[**which** keoidei gamjat dou daa].

WHICH they today also attend
'If they want promotion (in the match), only Group B and C may have lower marks than them, but they will also attend a match today.'

Indeed, (8) shows that no relative pronoun other than *which* is allowed in *which*-constructions. The use of *which* is insensitive to the head in the matrix clause and is free from the grammatical requirements observed in English.

(8) Only *which* can be used in *which*-constructions

a. *Ngo soeng faan [Hoenggong] [**where** Δ hai ngo zoengdaai ge deifong].
I want return Hong.Kong WHERE be I grow.up GE place
Int.: 'I hope to return to Hong Kong, a place where I grew up.'

b. *Ngo sik [Lei-lousi] [**who** Δ hai ngo jicin ge mentor].
I know Lee-teacher WHO be ngo former GE mentor
Int.: 'I know Mr. Lee, who is my former mentor.'

c. *Ngo sik-gwo [jat zung jyu] [**that** dak Jatbun sin jau Δ].
I eat-EXP one CL fish THAT only Japan first have
Int.: 'I have eaten a kind of fish that is only available in Japan.'

d. *Ngo tai-gwo [ni bun syu] [**whose** go zokze hai Hoenggongjan].
I read-EXP this CL book WHOSE CL author be Hongkonger
'I have read this book, whose author is a Hongkonger.'

All these cases would become possible if we changed the relative pronoun to *which*, despite mismatches in meaning between *which* and the antecedent. In other words, it is not that the whole paradigm of relative pronouns in relative constructions is borrowed from English, but rather, only a specific lexical element, i.e., *which*, is borrowed into Cantonese.

There are admittedly some variants other than the mere presence of *which* in *which*-RCs, such as *which is* and *in which*. Relevant examples are given below in (9). This observation is also noticed in previous works (e.g., K.-W. Leung 2010).

(9) Some variants of *which*

- a. Mui-nin dou jau hougei sau hit ge [**which is** Δ m-hai
every-year also have several CL popular GE WHICH IS not-be
jatsin coeng].
top.tier sing
(Lit.) ‘Every year there are several popular (songs), which is not sung by the top tier singers.’
- b. Jiwai wui hou aatjik-gam jyun [**in which** Δ jiging hou zeng].
thought will very oppressive-so finish IN WHICH already very good
(Lit.) ‘(I) thought (it) will finish in an oppressive way, in which (it is) already very good.’

Note that *is* in *which is* in (9) does not involve genuine agreement with a plural antecedent, since the head in the host is plural in number. In (12), the preposition-*which* sequence is also highly restricted and does not apply to other possible pied-piping cases in English such as *on which* and *by which*.

(10) No other variants are allowed

- a. *Aaming zidou ni-gin si [**on which** keoi jau m-tung ge jigin].
Aaming know this-CL event ONWHICH 3SG have different GE opinion
Int.: ‘Aaming knows this, on which he has different opinions.’
- b. *Ngo gei zungji baasi [**by which** ngo jatjat dap Δ faan hok].
I quite like bus BY WHICH I everyday take go school
Int.: ‘I quite like buses, by which I go to school everyday.’

Taking stock, these observations show that the choice of the linking word/relative pronoun in *which*-constructions is highly restricted: it requires the presence of *which*, and allows two and only two variants, namely, *which is* and *in which*, which allows no alternation and thus should be regarded as non-decomposable, lexicalized elements.

3.2. Genuine “gap”-less relative structures

It is not uncommon to observe in *which*-constructions that *which*-constructions can be used without a “gap”. For example, (7) involves a resumptive pronoun. It is also possible to have internal heads. In (11), the head *album* also appears as a co-indexing definite counterpart in the *which*-construction. As indicated in the English translation, such a property is not observed in English relative constructions.

(11) Internal heads in *which*-constructions

- Keoi giu ngodei tai album, [**which** go album zijau sap-gei zoeng soeng].
3SG ask us look.at album WHICH CL album only ten-several CL photo
(Lit.) ‘S/he asked us to look at album, which *the* album contained only several photos.’

Indeed, *which*-constructions can be merely semantically associated with the host clause. In these cases, *which*-constructions are genuinely “gap”-less. For example, in (12), the embedded subject *go singgwo* ‘the outcome’ can be construed as being semantically related to the matrix subject *heoi hok coenggo* ‘go learn singing’.

(12) “Gap”-less *which*-construction, case 1

Heoi hok coenggo zangkoeng zigei coenggo ge nanglik [**which** go
go learn singing improve self singing GE ability WHICH CL
singgwo zanhai hou minghin].
outcome really very significant
(Lit.) ‘(I) go learn singing to improve my singing ability, which the outcome is really significant.’

More importantly, there are also cases where the host clause and the *which*-RC are discourse-related, as if they are two separate clauses. In (13), it is impossible to identify the head in the host clause that is supposedly modified by the *which*-construction.

(13) “Gap”-less *which*-construction, case 2

Keoi m-hai hou sik cyulei ni di si [**which** keoi wui gei
3SG not-be very know handle this CL thing WHICH 3SG will quite
sitdai].
disadvantaged
(Lit.) ‘S/he doesn’t handle these things well, which s/he would be quite disadvantaged.’

Such cases are not unattested in English relative clauses. *Which*-constructions in Cantonese code-mixing speech have thus developed unique usages that are not typical of its alleged source of borrowing.

3.3. Unbounded dependencies

In cases where a head and a gap can be identified, they exhibit unbounded dependencies in several aspects. First, *which*-constructions allows a gap to be separated from the head across a finite/CP clause boundary. For example, the antecedent ‘simple report’ in (15) may be associated with a “gap” in the *which*-RCs across

(14) Cross-clausal dependencies

Honang jiu zou gaandaan bougou [**which** gojan gokdak [_{CP} Δ hai
Perhaps need do simple report WHICH personally think be
jiji bat daai]].
meaning not big
(Lit.) ‘(You) may need to do a simple report ... which (I) personally think that is meaningless.’

More importantly, the antecedent and the “gap” can span across typical syntactic islands (Ross 1967). For example, the gap in (15) is embedded in a complex NP island, whereas the gap in (16) is embedded in an adjunct island, but neither of them gives rise

to unacceptability. Island insensitivity suggests that the gap and the head are not related by syntactic dependencies (e.g., movement).

(15) Dependencies across complex NP islands

Ngo tung keoi dou zungji sik [laaulin] [which ngo jat zou zau teng-gwo
 I and 3SG also like eat durian WHICH I already then hear-EXP
 [NP keoi bei ngo gang zungji Δ ge gongfat]].
 3SG than I more like GE saying
 (Lit.) ‘We both like eating durians, which I already heard the saying that he like
 (durians) more than me.’

(16) Dependencies across adjunct islands

Sailou faangung zau jau [munzukungam] [which [Adj janwai keoidei faanhok
 youth at.work then have satisfaction WHICH because they at.school
 jatzik dak-m-dou Δ], soji keoidei tungsoeng dou hou jau dunglik].
 long gain-not-able so they usually DOU very have motivation
 (Lit.) ‘The young people at work usually get a sense of satisfaction, which because
 they cannot obtain (it) at school, they are usually self-motivated.’

On the other hand, K. W. Leung (2010:35) reports a case of co-construction, allowing cross-sentential usage of *which*-constructions. In (17), Speaker B utters a *which*-construction after Speaker A’s utterance, and the gap in the *which*-reconstruction is associated with the clause *jung-gwo jatci* ‘using it once’.

(17) Co-construction of *which*-constructions

- a. A: Ngo teng-gwo nei [jung-gwo jatci] wo3.
 I hear-EXP you use-EXP once SFP
 ‘I heard you using (it) once...’
- b. B: [which is Δ mou honang ge3].
 WHICH IS not possible SFP
 ‘...which is impossible.’

Also, clausal intervention is tolerated. The head and the gap can be intervened by another full clause, as shown in (18).

(18) Clausal intervention between the host clause and the *which*-construction

Keoi sik [saangcoi] si hou ginghei, man ngo dimgaai hang lok gam-dou
 3SG eat lettuce time very surprised ask me why will use that-much
 jau ge, [which keoi gokdak Δ houhousik].
 oil SFP WHICH 3SG think delicious
 ‘S/he was surprised when s/he ate lettuce, and asked me why I will use that much oil,
 which s/he thinks (it is) very delicious.’

Both island insensitivity and cross-sentential association are unlikely in English relative clauses. These observations suggest that the gap (if any) is not syntactically dependent on the host clause, and thus speak for a base generation approach to *which*-constructions.

3.4. The occurrence of SFPs

A final property concerns sentence-final particles (SFPs). Not only can *which*-constructions occur after SFPs, but they can also contain a different SFP than the host clause. This can be illustrated with (19) and (20).

(19) SFPs in *which*-constructions

- a. Nei di sausai m-gongzeng **wo5** [**which** nei-zigei jiging singjing-zo Δ
You CL skill not-clean SFP WHICH you-self already admit-PFV
laa1].
SFP

(Lit.) ‘Your skills are not good enough. Which you also admitted already.’

- b. Keoi zanhai hou zungsi keoi ge sijip **lo1** [**which** is Δ hai ngo
3SG really very treasure 3SG GE career SFP WHICH IS be I
hou jansoeng **ge2**].
very praise SFP

(Lit.) ‘S/he really treasures his/her career. Which is I really praise.’

Given that these SFPs in Cantonese occur in root clauses (Tang 2015), these cases suggest that the status of *which*-constructions is similar to the host clause in having a root clause syntax. Put differently, *which*-constructions enjoy a syntactic status that is substantially different from English relative constructions, the alleged source of borrowing (Chan 1993; K. W. Leung 2010).

4. Parenthetical meaning

In this section, I show that the proposition introduced by *which*-constructions is best characterized as parenthetical meaning. It displays the following four properties: (i) projective, (ii) illocutionarily independent, and (iii) discourse-new, and (iv) no binding.

4.1. Projective content

The proposition introduced in *which*-constructions is projective, and it escapes from the scope of various operators. For example, in (20), it is interpreted beyond the scope of an *if*-operator in a conditional sentence. The proposition in the *which*-construction contradicts the antecedent clause, but it does not give rise to infelicity.

(20) Projective from conditional sentences

- Jyugwo keoi hou jaucin, [**which** keoi m hai], keoi sat m wui
if 3SG very rich WHICH 3SG not be 3SG surely not will
gam zou.

so do.
 ‘If s/he is rich (which s/he isn’t), s/he would surely not do so.’

When embedded under intensional contexts with an epistemic agent, the propositions introduced by *which*-constructions are still oriented to the speaker, but not the matrix subject. For example, (21) involves an attitude report of the matrix subject *Aaming*. Within the intensional context, a *which*-construction introduces a proposition that contradicts the attitude report, i.e., Aaming’s worry would not be reasonable if he is committed to the proposition in the *which*-construction. However, (21) turns out to be felicitous. It still indicates that the speaker thinks that Peter would not be elected as the chairperson. The *which*-construction is interpreted beyond the scope of *daamsam* ‘worry’. In other words, the proposition introduced by *which* takes widest scope even when embedded under intensional contexts.

(21) Projective from intensional contexts sentences

Aaming daamsam jyugwo Peter zou-zo [wui zoeng], [which keoi m-wui
 Aaming worry if Peter be-ASP president WHICH 3SG not-will
 syun-dou], wui ling go zouzik mou-saai zicize.
 elect-able will make CL group lose-all supporter
 ‘Aaming worries that if Peter is elected president, which he will not be elected, will
 make the group lose all its supporters.’

Which-constructions also find a subtle difference with the clausal conjunction *ji* ‘and’ when embedded in *counterfactuals*. (22) below is modelled after Schlenker (2021). It involves a counterfactual event in the speaker’s future. However, only *ji* ‘and’ but not *which* can be used to introduce the proposition ‘he then called the Boss’.

(22) Projective from counterfactuals

[Context: someone made a big mistake at the department.]
 Jyugwo ngo tengjat daa-zo bei Aatau, [#which/ ^{OK}ji keoi ganzyu
 If I tomorrow call-ASP to Head WHICH and he then
 daa-zo bei Loubaan], gamzau daaiginsi.
 call-ASP to Boss then in.trouble.
 (Lit.) ‘If I called the Head tomorrow, #which/ ^{OK}and he then called the Boss, then
 (we are) in trouble.’

This contrast can be attributed to the fact that *which* involves matrix scope and fails to be interpreted under the counterfactual contexts. It is thus different from the clausal conjunction introduced by *ji* ‘and’. The contrast in embeddability is reminiscent of the one between clausal parentheticals and appositive relative clauses in English (cf. Schlenker 2021, 2023)¹, which in turn distinguishes *which*-construction from English appositive relative clauses.

¹ Schlenker (2021 p.10, emphasis mine) reports the following contrast between clausal parentheticals on one hand, and appositive relative clauses and conjoined clauses on the other (see also discussions in Schlenker 2023).

4.2. Illocutionarily independent

As already shown in (19), *which*-constructions can occur after SFPs of the host clause, and contain a different SFP, implying that they may be two root clauses. This point can be strengthened if we look at cases where the two clauses convey different illocutionary forces. For example, (23) shows that while the host clause is interrogative (as indicated by the question SFP *me1*, the *which*-construction is an assertion/exclamative sentence.

- (23) Mat nei mgeidak-zo keoi hai mou mat neoizai friend gaa3 me1?
 what you forget-PFV 3SG be not what female friend SFP Q
 [which is houdaai ge ginghou]!
 WHICH IS huge GE warning.sign
 ‘You forgot s/he doesn’t have female friends at all? It’s a huge warning sign (about the relationship).’

4.3. No binding allowed

An interesting property of parentheticals is that it consistently resists binding from the host clause. This can be illustrated with English cases in (24), which are judged unacceptable if the anaphors in the parentheticals (e.g., *as*-parentheticals, appositive relative clauses, and nominal appositives) are bound by the antecedent in the host clause.

- (24) No binding into English parentheticals (Potts 2005, p.80, 82)
 a. *No reporter₁ believes that, **as he₁ wrote**, Ames is a spy.
 b. *No reporter₁ believes that Ames is a spy, **which he₁ wrote in his column**.
 c. *No reporter₁ believes that Ames, **often the subject of his₁ columns**, is a spy.

Similarly, (25) shows that the reflexives in *which*-constructions cannot be bound by an antecedent in the host clause. Note that (25) is contrasted with (26), where binding into relative clauses is possible.

- (25) No binding into *which*-constructions
 *Wuizoeng doze soujau wuijyun₁, [which dou jung keoi₁ ge
 chairperson thank all member WHICH all use self GE
 fongfaat bong-zo-sau].
 way help-PFV-help
 (Lit.) ‘The chairperson thanked every member₁, which contributed in his₁ own way.’

- (26) Binding into relative clauses
 Wuizoeng doze [RC jung keoi₁ ge fongfaat bong-zo-sau ge]
 chairperson thank use self GE way help-PFV-help GE
 soujau wuijyun₁.
 all member
 ‘The chairperson thanked every member₁ who contributed in his₁ own way.’

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- (i) Context: *someone made a big mistake at the department.*
 a. If tomorrow I called the Chair, **who in turn called the Dean**, then we would be in deep trouble.
 ⇒ If I called the Chair, he would call the Dean.
 b. *If tomorrow I called the Chair (**he in turn called the Dean**), then we would be in deep trouble.
 c. If tomorrow I called the Chair **and he in turn called the Dean**, then we would be in deep trouble.
 ⇒ If I called the Chair, he would call the Dean.

The suggestion that the proposition brought along by *which*-constructions thus receives syntactic support from the resistance to reflexive binding.

4.4. Discourse-new, non-presuppositional

Our last argument for the parentheticality of *which*-constructions comes from a contrast with presuppositional meaning. *Which*-constructions introduce discourse-new information, and obey a triviality condition, which allows us to differentiate the propositional content introduced by *which* from presupposition. Precisely, a presupposition can be trivially true, as shown in (27). Although the facticity presupposition introduced by *zidou* ‘know’ in (27) is already part of the knowledge of the speaker, it is still felicitous.

(27)[Context: The speaker said that Mr. Wong is a linguist.]

Ji tunghok dou **zidou** Wong Lousi hai jat-go jyujinhokze.
and student all know Mr. Wong be one-CL linguist
‘And all students know that Mr. Wong is a linguist.’

However, when the proposition introduced by *which*-constructions is trivially true in (28), it results in infelicity or give a sense of redundancy.

(28)[Context: The speaker said that Mr. Wong is a linguist.]

#Dong geize fongman [Wong Lousi], [**which** Δ hai jat-go jyujinhokge],
when reporter interview Mr. Wong WHICH be one-CL linguist
geize ge taaidou hou jausin.
reporter GE attitude very friendly

(Lit.) ‘When the reporter interviewed Mr. Wong, which is a linguist, the attitude of the reporter is pretty friendly.’

One more difference with presupposition can be observed in the strength of projection. As argued in Koev (2022), parenthetical inferences project more strongly than presupposition. For example, the projection of presupposed content can be blocked if it is entailed by the conditional antecedent, as illustrated in (29).

(29) Blocked projection of presupposition (Koev 2022:67)

a. Hillary regrets she kissed Obama.

(presupposed: *Hillary kissed Obama*)

b. If Hillary kissed Obama, then she regrets she kissed him.

(Not presupposed: *Hillary kissed Obama*)

A structurally parallel counterpart of (29b) with *which*-constructions shows a sharp contrast. The infelicity of (30) can be accounted for if the proposition introduced by the *which*-constructions remains strongly projective. Particularly, when it projects through the conditional operators, the speaker is certain about the proposition, but at the same time, the conditional antecedent also entails the same proposition, rendering the speaker being both certain and uncertain as to whether a teacher is respectable. (30) is thus

infelicitous. The same infelicity is not observed with (29b), as the projection of the presuppositional meaning is blocked.

(30) No blocked projection of *which*-constructions

Jyugwo lousi wui saujan-zyungin, Aaming soeng zou lousi,
 if teacher will be-respectable Aaming want be teacher
 [**which** wui saujan-zyungin].
 WHICH will be-respectable
 ‘If a teacher is respectable, Aaming wants to be a teacher, which is respectable.’

Given these differences, the nature of the propositional content in *which*-constructions cannot be reduced to presuppositional meaning, indirectly supporting the suggestion that they introduce parenthetical meanings. In the next section, I will turn to an analysis on *which*-constructions.

5. Proposal: *which* borrowed as an overt force operator

5.1. Two ingredients

Taking stock, the desiderata of an analysis on *which*-constructions can be summarized in Table 1 below.

Syntactic properties	Semantic properties
(a) Only <i>which</i> can be used.	(e) Projective content is introduced.
(b) Genuinely “gap”-less cases are possible.	(f) They are illocutionarily independent.
(c) Dependencies are unbounded.	(g) No binding into the constructions.
(d) They allows an independent SFP.	(h) They introduce discourse-new content.

Table 1 – the syntactic and semantic properties of *which*-constructions

There are two ingredients in my proposal. First, I argue against a modification account on *which*-constructions, and I propose that *which*-constructions are clausal parentheticals, where the parenthetical meaning is introduced by *which*. The status of *which*-constructions is similar to English clausal parentheticals as in (31), except that there is no overt marking in English.

(31) English clausal parentheticals (Blakemore 2006:1671)

Wo do not need to begin with the children – [they will follow] – but with the adults.

Second, following the illocutionary approach to parentheticality proposed in Koev’s (2022), I suggest that *which* is lexically borrowed from English as an overt force operator for parenthetical meanings. The semantic contribution of *which* is inherited largely from the relative pronouns in appositive relative clauses in English. However, it departs from the syntax of relative pronouns and show no signs of medication. The proposal amounts to the suggestion that there is no syntactic/constructional borrowing at all (*pace* Chan 1993, 2022, K. W. Leung 2010), but only a case of lexical borrowing of *which*.

Before I detail my proposal, it should be noted that K.-W. Leung (2010:40) mentions in passing that *which* in the relevant constructions can be replaced by *ji* ‘and’ in Cantonese.

The intuition is on the right track, but the example in (22) suggests that *which* and *ji* ‘and’ are not always interchangeable. On the other hand, Chan (2022:7) also mentions that *which* serves as “a discourse or pragmatic marker” whose function is to introduce “a personal assessment on a situation or entity expressed in the first/matrix clause.” My parenthetical treatment on *which* rules in this usage, if the proposition is about one’s assessment, but it need not always be the case. As exemplified in (1b), the proposition in the *which*-construction is not about the speaker’s assessment, but a report of an objective event.

5.2. Details of the proposal and an illustration

Regarding the internal syntax of *which*-constructions, I propose that *which* is a functional element that resides in a Force head, under a cartographic understanding of clausal structures (Rizzi 1997, *et seq.*). It is a force operator specifically for the introduction of parenthetical, declarative proposition, as opposed to a root force operator (Koev 2022). The structure of a *which*-construction can be schematically represented in (32).

(32) The structure of a *which*-construction

... [ForceP WHICH [TopicP *pro*_i [FocusP ... [FinP S V O/*eci*]]]]

Note that the semantic/discourse linkage between a *which*-construction and the host clause is established via a null topic *pro* residing in the Topic projection, whose referent depends on the discourse contexts. It may refer to a salient referent established in the host clause, i.e., the cases with heads (e.g., (1b)), or a salient event or proposition in the discourse, i.e., the cases without heads (e.g., (1a)).² The flexibility of the semantic relation between the two proposition is mediated by the topic *pro*.

Turning to the semantic contribution of *which*, I adopt Koev’s proposal that parentheticals are introduced by a force operator, as defined in (33). Essentially, *which* introduces a propositional referent, restricts its value to the content of its prejacent, and asserts its content.

(33) The lexical semantics of *which*

- a. $\llbracket \textit{which} \rrbracket = \lambda p. \text{P-DECL}^p c(p)$
- b. Parenthetical operator (P-DECL) is defined as:
 $\text{P-DECL}^p c(p)$ for $\exists p \wedge \phi_p \wedge c + p$
 $(c + p \text{ commits the speaker of (the referent of) context } c \text{ to the proposition } p)$

Crucially, *which* is different from the non-parenthetical, root force operator in two ways. First, *which* is an overt realization of a force operator for declaratives. Second, the propositional referent introduced by *which* need not be at-issue (Koev 2022; see section 6.2. for discussions). At-issueness is a signature property of a root force operator, rather than a parenthetical force operator, as defined in (34). $c \oplus p$ requires that the proposition expressed by *p* be relevant to the question under discussion, hence at-issueness.

² The suggestion that a null *pro* in Chinese can refer to both entities and events/propositions is not new in the literature. See, for example, discussions on sluicing-like constructions in Mandarin (e.g., Wei 2004).

(34) Root operator (R-DECL) is defined as:

$$\mathbf{R-DECL}^p_c(\varphi) \text{ for } \exists p \wedge \varphi_p \wedge c \oplus p$$

I illustrate the proposal with the example in (1b), repeated below in (35).

- (35) Keoi gin-dou bou dinwaa [which keoi waa soeng mai Δ] wo5.
 3SG see-ASP CL telephone WHICH 3SG say want buy SFP
 ‘He has seen a telephone which he says he wants to buy.’

Under the current proposal, (35) has the structure in (36), which involves two distinct Force phrases, headed by different Force head. In ForceP1 (i.e., the host clause), $\emptyset$ is a root force operator, whereas *which* is a parenthetical force operator. The two ForcePs are semantically related by the *pro* in ForceP2, which co-indexed with a salient referent in ForceP1, i.e., phone. The empty category (*ec*) in *which*-constructions is bound/controlled by *pro*.³ The semantics of (35) is sketched in (37), where only ForceP1 is obligatorily at-issue. ForceP2 is unspecific for at-issueness.

(36) Structure for (36) (in English glosses for readability)

$$[\text{ForceP1 } \emptyset [\text{FinP he saw phone}_i] \\ [\text{ForceP2 WHICH } [\text{TopicP } pro_i [\text{FinP he says want buy } ec_i]]]]]$$

(37) A sketch of the semantics of (36)

- a. $[\text{ForceP1 } \mathbf{R-DECL}^p_c [\text{FinP he}^x \text{ saw phone} \\ [\text{ForceP2 } \mathbf{P-DECL}^p_c [\text{FinP he}^x \text{ says want buy }]]]]$
 b. $\exists p \wedge \text{see}_p(x, \text{phone}) \wedge c \oplus p \wedge \\ \exists q \wedge \text{say}_q(x, \text{want.buy.phone}) \wedge c + q$

Under this construal, *which*-constructions do not involve any modification structure. Instead, *which* lexically borrowed as a marker for clausal parentheticals in code-mixing speech in Cantonese, and the clause within a *which*-construction involves a (canonical) predication relation between a null topic and its comment.

5.3. Accounting for the properties of *which*-constructions

A few words need to be said on how the proposal derives both the syntactic and semantics properties of *which*-constructions. Table 1 is repeated below for convenience.

Syntactic properties	Semantic properties
(a) Only <i>which</i> can be used.	(e) Projective content is introduced.
(b) Genuinely “gap”-less cases are possible.	(f) They are illocutionarily independent.
(c) Dependencies are unbounded.	(g) No binding into the constructions.
(d) They allow an independent SFP.	(h) They introduce discourse-new content.

Table 1 – the syntactic and semantic properties of *which*-constructions (repeated)

³ The empty category may be derived via Argument Ellipsis, the precise nature of which is not directly relevant to the proposal.

Regarding the syntactic properties, (a) follows from the fact that *which* and only *which* is lexically borrowed into Cantonese. The rest of the whole paradigm of relative pronouns in relative constructions in English are not involved. For (b), if there is no gap at all, like the cases discussed in section 3.2., the topic can be realized overtly as in (11). Alternatively, the proposition may be predicated on the null topic *pro*, instantiating a topic-comment structure, which is commonplace in Chinese languages, e.g., (12) and (13). It comes as no surprise, then, (c) follows, since there is no movement in *which*-construction at all. A base generated *pro* is postulated in the topic position. Lastly, since *which* heads a ForceP, it is thus expected that it can host SFPs which are in various positions in the CP periphery (Sybesma & Li 2007, i.a.).⁴

As for the semantic properties, (e), (f) and (h) are direct consequences of suggestion that *which*-constructions introduce a ForceP that is distinct from the ForceP in the host clause. The *which*-headed ForceP perform a distinct illocutionary force, and should not be conflated with presuppositional meanings, as it is discourse-new. As for (g), I follow Koev (2022) in assuming that each ForceP must be syntactically closed for quantifier binding and they cannot contain any unbound pronouns, as stated in (38).

(38) No Free Pronouns (Koev 2022:88)

All bound pronouns must contain a quantificational antecedent within the same Force Phrase.

In sum, the current proposal derives the relatively syntactically independent nature and the parenthetical meaning of *which*-constructions. In the next section, I briefly discuss some further supporting evidence to this proposal.

6. Supporting evidence

6.1. No interrogative parentheticals

The clausal parentheticals introduced by *which* comes with a restriction that is not observed in English. In particular, clausal parentheticals in English can be interrogative, as shown in (39).

(39) Interrogative parentheticals in English

The drive – **what as his name again?** – had disappeared. (p.c. *****)

However, the Cantonese counterpart of (39) in (40) shows that *which* fails to introduce an interrogative parenthetical. The parenthetical in (40) is a *wh*-question, containing *me* ‘what’ and the echo question particle *waa2* is also incompatible with *which*. By contrast, in the absence of *which*, the same sentence is acceptable, as in (41).

(40) No interrogative parentheticals introduced by *which*

*Go sigei – [**which** keoi giu me meng (waa2)?] – datyin siusat-zo.
CL driver WHICH 3SG call what name Q suddenly vanish-PFV
(Lit.) ‘The driver – which what is his name – vanished suddenly.’

⁴ An exception is however that question particles are disallowed in *which*-constructions, a point to which we return shortly in section 6.1.

(41) Interrogative parentheticals introduced by a null Force head

Go sigei – [ø keoi giu me meng (waa2)?] – datyin siusat-zo.
CL driver 3SG call what name Q suddenly vanish-PFV
(Lit.) ‘The driver – which what is his name – vanished suddenly.’

The contrast between (40) and (41) follows naturally, as the declarative requirement is part of the lexical specification of *which*, which is a *declarative* force operator. The null counterpart is less specified and can be ambiguous between a declarative one and an interrogative one.

6.2. Non-obligatory at-issueness

The illocutionary approach to parentheticality advocated in Koev (2022) suggests that not-at-issueness is not a hardwired property of parentheticals. Instead, he proposes that parenthetical clauses are left open with regard to at-issueness. It is the root clauses that must be at-issue. This construal of parentheticality makes two predictions on *which*-constructions. First, it is expected the proposition in *which*-constructions can be targeted for denial. This is borne out in (42) and (43). In (42), the proposition *they are still having the main course* is introduced as a parenthetical. Crucially, it can be denied by the continuation in (43a). Note that the host clause can also be denied by (43b).

- (42) A: Cyun-coeng dak jat-zoeng toi hai ji-jan-toi, [which keoidei
whole-venue only one-CL table be two-person-table WHICH they
zung sik-gan zyucoi].
still eating main.course
(Lit.) A: ‘There is only one table for two people in the venue, which they are still
having the main course.’

(43) Both the *which*-clause and the host clause can be denied

- a. B: M-hai aa3. Keoidei sik-gan timban. (Denying the *which*-clause)
not-be SFP they eat-PROG dessert
B: ‘No, they are having deserts.’

- b. B: M-hai aa3. Go-dou jau leong-zeong toi. (Denying the host clause)
not-be SFP there have two-CL table
‘No, there are two such tables.’

Second, *which*-constructions can also serve as an answer to a question, as shown in (44). It can thus be concluded that clausal parentheticals are indeed open for at-issueness, despite being not-at-issue in most cases. Rather, it is the root clause that must be at-issue. Such a behavior of *which*-constructions lends further support to Koev’s proposal.

(44) *Which*-constructions can serve as an answer to a question

[In response to the question “*Why don’t you say a word?*”]
Gin si houfukzap, [which ngo mou me zingsan heoi gong].
CL matter complicated WHICH I not.have what energy to discuss
‘The matter is complicated, and I don’t have any energy to discuss it.’

6.3. The clause size of *which*-constructions

Last but not least, given that *which* introduces a Force Phrase, it is predicted that the clause can host elements in the CP periphery. This prediction is borne out. For example, a Focus projection in between a ForceP and a FinP is allowed, as shown in (45). The *which*-construction can contain a Focus construction involving *lin* ‘even’ (Lee 2022, 2024).

- (45) Haa-jat-bou zauhai maai daancong, [**which** [_{FocusP} lin daancong
 Next.step be buy jumping.bed WHICH even jumping.bed
 dou jau review]].
 also have review
 (Lit.) ‘The next step is to buy jumping bed, which even jumping bed has reviews.’

Additionally, speaker-oriented adverbs, presumably heading the EvalP in the CP periphery, can appear within *which*-constructions as well.

- (46) Di namjan ciudaai dungzok fing sau, [**which** [_{EvalP} houzoi
 CL.PL man great.extent motion wave hand WHICH luckily
 keoidei mou lo-zyu penlight]]].
 they not.have hold-DUR penlight
 (Lit.) ‘The men are waving their hands at full, which luckily they are not holding any penlight.’

7. Concluding remarks

In this paper, I discussed empirical properties of *which*-constructions in code-switching speech in Hong Kong Cantonese. I have discussed how they deviate from a relative construction and the parenthetical nature of the proposition introduced in these constructions. Based on these observations, I argued for a clausal parenthetical analysis on *which*-constructions following Koev’s (2022) illocutionary approach to parentheticality. I suggest that they instantiate a ForceP distinct from the host clause, and they do not involve modification, but rather predication. As such, *which* is lexically borrowed as an overt Force operator that overtly marks parenthetical clauses.

There are a few implications. First, there is no evidence of syntactic borrowing in Hong Kong Cantonese code-mixing speech, at least not from *which*-constructions. The relative pronoun *which* is lexically borrowed, with considerable degree of semantical bleaching or grammaticalization, and serves as a functional morpheme that introduces a topic-comment structure (*pace* K. W. Leung 2010; Chan 2022). Second, neither English-style relative structures nor post-modification gets into Cantonese grammar. It does not defy the Matrix Language Frame Model (Myers-Scotton (1993, 2002)), *contra* Chan (2022). Third, to the best of my knowledge, no parenthetical meaning is marked grammatically in Cantonese. Thanks to lexical borrowing from English, *which* represents the first overt marker of parenthetical meaning in the language. Lastly, to the extent that Koev’s analysis can be applied to the clausal parentheticals in Cantonese, there seems no need to invoke the two-dimensional approach to parentheticals, as suggested in Potts (2005). A unidimensional semantics can be maintained (Schlenker 2021, 2023).

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